NG5001
Academic Communication for Graduate Researchers

Key points from Week 2's tutorial discussions

- This week's lesson addressed the critical dispositions expected of researchers while reading for research, warranting claims, communicating ideas in cross-disciplinary contexts and establishing a research niche
  - ✓While different disciplines warrant claims differently, adequate and appropriate warranting logically leads to the conclusion (from online task in Week 1)
  - ✓We set our own agenda when reading and reviewing literature; it is this that makes our response original, critical and more nuanced
- We also briefly discussed the challenges and concerns surrounding cross-disciplinary communication and collaboration
  - ✓The need to establish a common ground
  - ✓The need to be explicit about disciplinary norms and assumptions

To collaborate and communicate across disciplines, it helps if we understand the different research paradigms.

- These are worldviews (i.e., disciplinary assumptions, values, concepts and practices) that shape how we think about research
- Major paradigms include:
  - ✓Positivism that focuses on objectivity, causality & generalizable laws; testing of hypotheses, quantitative methods and empirical evidence are valued
  - ✓Interpretivism/ constructivism that foregrounds subjectivity, human agency and positionality in specific social contexts; reality is perceived as co-constructed hence qualitative methods are preferred to capture the richness and complexity of human experiences
  - ✓Critical/ transformative paradigm investigates issues such as social inequality and hegemony with the aim of bringing about social change; critical theory is central to the analysis, and such a paradigm is open to interdisciplinary approaches

Although their philosophical foundations differ, modern day complexities have given rise to mixed-method research.

> "… these paradigms have not existed in isolation but rather have influenced and interacted with one another. Researchers from different paradigms have engaged in critical dialogue, borrowing concepts and methodologies from one another. This interdisciplinary exchange has led to the emergence of mixed-methods research, which seeks to integrate quantitative and qualitative approaches, recognising the complementary strengths of each paradigm. Therefore, the historical development of research paradigms reflects the dynamic nature of knowledge production in the social sciences" (Omodan 2025).

Understanding the distinctions and compatibilities of different paradigms enhance criticality in research.

- Encourages researcher reflexivity, for e.g.,
  - ✓Despite the focus on objectivity, researchers adopting the positivist approach recognizes the **inherent bias** in the choices they make and the assumptions they hold
  - ✓Researchers adopting the interpretivist approach makes clear the **situatedness** of the findings and accounts for **how their presence may have an impact** in participants' responses or behaviours during interviews and participant observations
  - ✓Researchers who adopt the mixed method approach incorporates a quantitative element into their research design for purposes of triangulation to mitigate the subjectivity in their research to **ensure robustness and validity** while justifying how the qualitative element adds **nuance** to their findings/ discussion
- Encourages dialogue and collaboration for better research outcomes

Different disciplines warrant their claims differently.

(cf. Task 4 Unit 1.2)

Type of research: Interpretive, contextual & human-centred

Warranting in such a research can include:

- Quantitative method (surveys, standardized tests, statistical analysis)
- Quantitative claims rely on validity, reliability, and generalizability
- Qualitative method (e.g., case studies, classroom observations, interviews, ethnographies)
- Qualitative claims rely on credibility, transferability, dependability, and confirmability
- Use of theoretical framework to interpret data & support claims
- Applicability & impact in real world education settings

Type of research: Empirical, experimental, reductionist

Warranting in such a research can include:

- Controlled experiments
- Statistical significance
- Peer-reviews
- Replication
- Biological plausibility and molecular mechanisms to support finding
- If involving human subjects, IRBs

Type of research: Applied, technical & solution-oriented

Warranting in such a research can include:

- Mathematical/ computational modelling to predict
- Prototyping/ testing
- Established standards & specifications
- failure analysis & safety margins
- Peer reviews & patents

Examples in Task 4 come from the fields of education, biology/ medicine & engineering with their typical ways of warranting mentioned here.

We do research to:

Advance knowledge

Advocate for a cause (a civic duty)

Bring about societal impact

*All these require **effective communication** especially the discursive & strategic competences to **persuade** the audience on the significance of the research.*

As part of creating a research niche, a literature review should be critical, nuanced and reflexive.

1. Decide on the works that matter to the research
2. Determine the appropriate structure of the review, for e.g., thematic, conceptual, chronology, or methodological depending on how your research fits in
3. Cast these works as a conversation or narrative that leads to your research as the solution/ answer to a problem
4. Evaluate the relevance, quality and reliability of the works that frame your research
   - ✓ Use evaluative words, boosters and hedges to strategically foreground key claims or express caution
5. Have a clear gap statement that logically follows from the presentation of the lay of the field

*A literature review should **not** be summaries of all the works in the field. (cf. Task 5 Unit 1.2)*

These 3 examples of writing from different disciplines show how research niche is established.

(cf. Unit 1.2 Task 7)

Task 7: How scholars establish their niche

Read the following peer-reviewed article introductions (Texts A-C). How do the authors establish their niche? Pay close attention to how the literature review contained within helps to position the research within a given field or topic. Also pay attention to how their *language* helps to establish the importance of their niche.

Para 1: Broad intro then narrows down to US context and mentions the importance of topic - " annual fatalities" i.e., Structure: General-Specific

Text A Quantitative but language us persuasive and urgent; not 'neutral' in tone.

There are an estimated 1.2–5.5 million snakebites annually around the world, with approximately 20,000–94,000 attributed annual deaths [1]. The annual cost burden of snakebites is significant, and worldwide morbidity and mortality represent a neglected global public health problem [1,2]. In the United States, there are approximately 7,000–10,000 annual emergency department visits from snakebites [3], with 32–60% from venomous species, the majority being pit vipers [3,4], and 5–7 annual fatalities per year [3-5].

Snakes are poikilothermic animals whose activities, life cycle, and geographic distribution are closely tied to ambient temperature [7,8], and vary as a result of climate change [9]. There is at minimum a 99% chance that average global temperatures have significantly increased since 1950 to present times [11], and this global warming or "climate energizing" frequently results in increased extreme weather patterns [12,13].

specific context

language is emotive emphasizing urgency

Para 3: Niche - the relationship between snakebites & climate change (implying originality of research) Geographical location as the gap

These fluctuations of extreme weather (both drought and high precipitation) impact human health through an increase of natural disasters such as intense heat waves, droughts, and coastal flooding [14]. Likewise, climate change in North America is associated with changing distribution of venomous species that may lead to increased human morbidity [12,15,16]. Climate change has been described as the greatest global threat of the twenty-first century [17], but the effect on snakebites in North America has not been examined.

The topic sentence, however, needs to be changed especially when the focus is on how droughts correlate with snakebites.

There is some evidence for increased incidence of snakebites with warming temperatures, however interpretation is limited to Central American meteorological patterns [9], and the current literature on snakebites in the United States provide limited insight into this relationship [3,18]. With the recent most severe drought and precipitation seasons ever recorded in California [19], we sought to correlate the relationship of climate trends with snakebite incidence in California over 20 years. The study's primary hypothesis was that the severity of drought would correlate with increased incidence of snakebites and could be predicted by weather patterns.

Para 4: Niche is further refined to specific context - California.

Reviewed literature shows how available research is limited, justifying the need for this study; the research aims & hypothesis then logically follow.

Source: Adapted from Phillips C., Lipman G., Gugelmann H, Doering K., & Lung D., "Snakebites and climate change in California, 1997-2017," Clinical Toxicology, 2019.

Strong impact: Scientific study with quantitative data but writing still connects human experience and societal relevance.

Structure: broad to specific to broad again (impact)

Qualitative using narrative, case studies & descriptive data.

Rationale through a narrative & a story instead of announcing a research gap - but the gap is still clear.

Text B

Para 1: broad claim that also hints at complexity; also the area being insufficiently studied

Indigenous peoples make up over 5% of the world's population, their territories comprise at least 20% of the land area of the planet, and these lands host an estimated 80% of the world's biodiversity (Stevens 2014). The relationships between indigenous land tenure and biological diversity is a challenging area of research, but several studies suggest that indigenous peoples' place-based values, institutions, and practices help promote biodiversity.

Indigenous lands often support higher native and rare species richness (Redford and Stearman 1993, Peres 1994, Arcese et al. 2014) and lower rates of deforestation (Nolte et al. 2013, Ceddia et al. 2015). However, beyond a handful of studies looking at biodiversity and land use change, few researchers have looked at the broader ecological outcomes of indigenous land tenure.

Para 2: move to specific indicators; research gap

Para 3: Narrative through case studies
Lit review adds credibility & significance through humanizing the research; reader convinced of value of research

Indigenous peoples in Wisconsin manage forestlands and wildlife by merging professional standards of forestry and wildlife practice with their own culturally specific traditional ecological knowledge and land-based values. Although commercial and ecological forestry best-practices are utilized in tribal forestry programs, indigenous communities' own knowledge systems and values have a strong influence. This has often led to distinct forest and wildlife management goals, institutions, and practices (Trosper 2007, Dockry 2012, Reo and Whyte 2012). Indigenous forestry in the United States and Canada has several unique aspects (Bengston 2004). The Cherokee Nation of Oklahoma, for example, plants a diverse suite of culturally significant tree species and favors nonchemical methods of land clearing that promote the growth of food and medicinal plants while protecting harvesters' health (Carroll 2015). Anishnaabe of Shoal Lake in Northwestern Ontario use succession management via landscape burning and timber harvest to promote biodiversity and support local livelihoods (Berkes and Davidson-Hunt 2006).

Para 4: establishes the problem, hints at tension

Indigenous Nations and nonindigenous forest and wildlife managers generally profess similar goals typically centered around sustaining resource values and yields, biological diversity, and ecosystem productivity despite some differences. Do these shared goals lead to similar outcomes on the lands managed by these two groups? Here, we explore that question within the specific context of northern Wisconsin where a mix of private industrial, private nonindustrial, and county, state, and national public forestlands surround four major "Indian reservations," i.e., lands under jurisdiction of Indigenous Nations in the U.S. (Fig. 1A).

Research is exploratory - cf. Text A

Para 5: highlights the contribution; clarifies research aim

This study thus fits within existing literature concerning the foundations and outcomes of indigenous forest and wildlife management. Our focus here, however, is to quantify the ecological outcomes of forest and wildlife management and links between the two. That is, have the land-based values and practices of indigenous peoples in Wisconsin led to significantly different ecological conditions on their forestlands relative to neighboring lands? In the discussion, we return to the question of how differences in values and goals contribute to the differences in outcomes we document.

Source: Adapted from Waller, D., and Reo, N., "First stewards: ecological outcomes of forest and wildlife stewardship by indigenous peoples of Wisconsin, USA," Ecology and Society, 2018.

Lit review - establishes a narrative of ecological stewardship through the case studies of different indigenous cultures (humanizes them), acknowledges what has been done, points out what has not be sufficiently explored, positions study as exploratory and not hypothesis-driven.

Functional, focuses on problem-solving that typifies engineering research

Text C

Structure: Gen prob - current limitations - justification of new method

Para 1: broad prob before introducing the narrowed-down focus
Urgency signalled through langage - "must", "a very important stage"

For wind energy to become competitive with respect to other sources of energy, the initial consideration must be to reduce the cost of energy from wind power. In modern wind power research, how to minimize the cost of a wind turbine per unit of energy is an important task. The shape of the rotor blades plays a decisive role in determining the overall aerodynamic performance of a horizontal axis wind turbine (Fuglsang, 1995; Giguere and others, 1999; Benini, 2002). Thus, aerodynamic optimization of the blade shape is a very important stage in the design and manufacturing of wind turbines.

In earlier days, the Glauert and Wilson methods were mostly used for blade design (Spera, 1994). The objectives of these methods were to obtain the maximum power coefficient of each blade section at the design wind speed. Because the time variation characteristics of wind speed are not taken into account, blades designed by these methods cannot achieve the maximum annual energy output. Furthermore, design results from these methods must be substantially corrected to get smooth chord length and twist distributions (Spera, 1994). Because corrected results already deviate from the design points, effectively controlling design results poses problems. Conventional search algorithms, such as the feasible direction method and the complex method, are often prone to converging on the local optimal point (Gen, 1999). For some complicated problems, it is difficult to obtain a global optimal result.

Para 2: Lit review of classical models provides brief history but framed to show their limitations through word choice.

Para 3-4: Summarizes related research & presents own research as an extension

Several studies have been carried out on the viability of the wind turbine for power generation. Grauers, (2003), worked on efficiency of three wind energy generator systems. Jureczko and others, (2005), developed optimization of wind turbine blades. Afolami, (2007), improved the operation of a wind turbine through replacement of some components of the wind turbine. Michael, (2007), showed that modern wind turbines have become an economically competitive form of clean and renewable power generation. In his study, he carried out an economic optimization analysis of a variable speed, three blade, and horizontal-axis wind turbine.

This work presents design tool for optimizing wind turbine blades. This considers optimizing the blade of wind turbines with respect to maximizing the energy yield of a wind turbine. To achieve the optimization in this work, the genetic algorithm code in MATLAB is written for the turbines and this code can be used for any type of wind turbine provided that the objective function is the same.

Source: Adapted from Eke, G. B., & Onyewudiala, J. I. (2010). *Optimization of wind turbine blades using genetic algorithm*. Global Journal of Researches in Engineering: Mechanical and Mechanics Engineering, 10(7), 20–29.

Typical STEM writing - clarity & efficiency are valued (conciseness)
Use of language is still strategic, persuasive - clear positioning, confident & strong authorial stance

Follow-up activity:
Search for a leading journal in your field, then use relevant keywords to find three research articles related to your interests. Review their introductions, focusing on how the authors position their work in relation to prior studies. How do their approaches compare to the examples above?